KWON YEWON

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RESEARCH INTERESTS

- Computer Graphics and 3D Reconstruction of Real-World Spaces
- Generative AI for 3D and Visual Content (Diffusion Models, LoRA Fine-Tuning)
- XR and Interactive Media

EDUCATION

Mar. 2022–Feb. 2027 **Duksung Women's University, Seoul, South Korea**
B.S. in IT Media Engineering (Expected Feb. 2027)
GPA: 4.1/4.5 | Ranked 1st in Department (Fall 2023, Spring 2026)

PUBLICATION

2025 Youngho Kim, Junghyun Han, **Yewon Kwon**, Yeonwoo Kim, and Dahee Yoo. **"A Study on the Generative AI Reproduction of Traditional Folding Screens: Focusing on Digital Folding Screen."** The International Journal of Advanced Smart Convergence, 14(4), 305–312, 2025. doi:10.7236/IJASC.2025.14.4.305

RESEARCH EXPERIENCE

Mar.–Dec. 2025 **Undergraduate Researcher, Interactive eXperience Research Lab (IXR Lab)**

- Prepared and de-duplicated minhwa landscape images with CLIP similarity; refined ViT-generated tags for a 70-image research dataset.
- Configured Stable Diffusion v1.5 LoRA training and evaluated sampling-step and CFG-scale combinations.
- Contributed to exhibitions: Wind and Folding Screen (Art Korea Lab, Sep. 2025), Digital Folding Screen Showcase (Jeonju Univ. X Space, Nov. 2025).

SELECTED PROJECTS

Mar. 2026–Present **V-O — Spatial Analysis Platform from iOS Indoor Scans to VR** (Capstone)

- Built a platform that turns iOS indoor scans into editable VR spaces; implemented floor-plan-based furniture-placement optimization.
- Used Claude API to generate human-like layout constraints as auxiliary constraints for the optimizer, while collision, distance, and rotation stay rule-based.
- Deployed and operated the optimization pipeline as FastAPI services on AWS (EC2, RDS, S3) with GitHub Actions; built a hybrid-search RAG (BGE-M3, pgvector) for design rules.
- *Explored but not adopted:* 3D Gaussian Splatting from room videos (COLMAP SfM), 16-view RGB-D rendering of RoomPlan scans for SAMPart3D, and SAM → LaMa → Hunyuan3D-2.1 object-to-3D; inpainting filled gaps but did not recover occluded furniture shapes.

Jun.–Aug. 2025 **Hwagyeong / Wind and Folding Screen — Interactive Media Art** (with MO-ON Company)

- Collected and preprocessed ~40 minhwa and landmark images; co-trained Stable Diffusion LoRA and compared strength values.
- Built a Unity (C#) touchscreen app where users select and color 3D animals that appear on a generative folding-screen display in real time, linked via UDP with acknowledgments.

May–Jun. 2026 **FingerType — ASL Recognition Web App**

- Retrained YOLOv8n and compared six dataset versions; adding self-collected hand images raised validation mAP50 from 0.946 to 0.968. Delivered real-time webcam and video inference.

Jul.–Aug. 2026 **Mediary — AI Avatar Diary Service**

- Built an evidence-grounded LLM diary pipeline with multimodal input and emotion verification.

AWARDS AND CREDENTIALS

Nov. 2025 **Excellence Award, College of Science and Technology Academic Festival**
Duksung Women's University (three-person team)

Sep. 2024 **SQL Developer (SQLD)**
Korea Data Agency

TECHNICAL SKILLS

Programming	Python, C# (Unity), Java, SQL | FastAPI, PostgreSQL, AWS (EC2, RDS, S3), Git, GitHub Actions
3D & Vision	3D Gaussian Splatting, COLMAP, SAM, Hunyuan3D-2.1, Blender (Python), YOLOv8, MediaPipe, OpenCV
Generative AI	Stable Diffusion LoRA, LaMa inpainting
LLM & Data	Claude API, CLOVA Studio, RAG (BGE-M3, pgvector), Neo4j